EXP.NO:03

PROBLEM 3 : FIND THE TIME COMPLEXITY USING COUNTER METHOD

PROBLEM STATEMENT:

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

CODE:

Answer:

```c
#include<stdio.h>
int main(){
    int n, count =0;
    scanf("%d", &n);

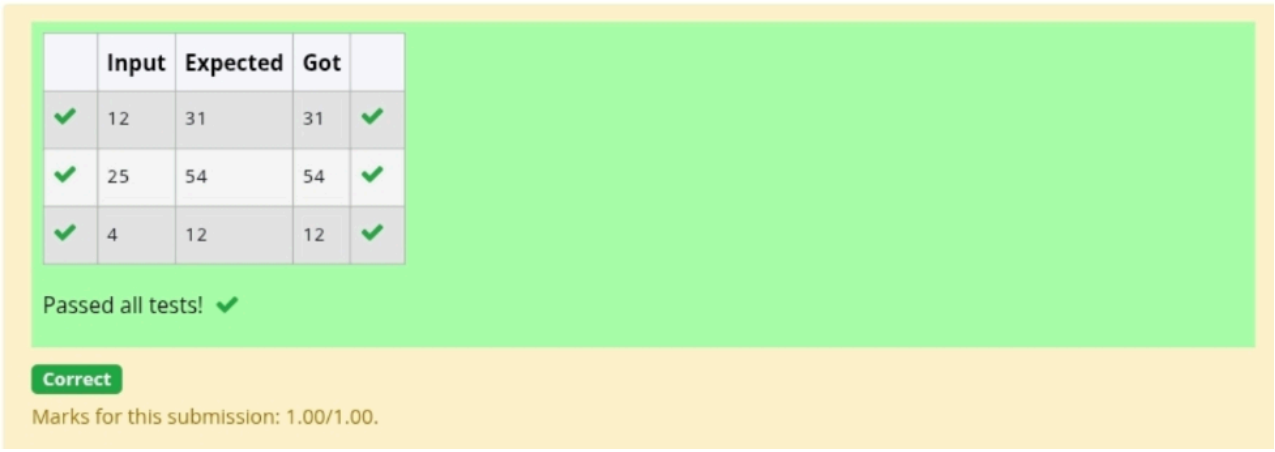
    for(int i=1;i<=n;i++){
        count++;

        if(n%i==0){
            count++;
            }
        count++;


    }
    count++;
    printf("%d", count);
    return 0;
}
```

EXP.NO:03

OUTPUT:



	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.